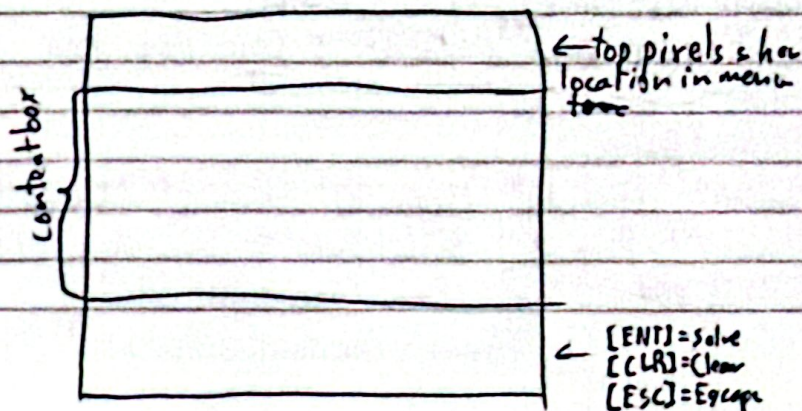


Projects

1. Pocket Formula Solver (4-6 weeks)

- has a database of physics, engineering and math formulas, plug in known formulas and will use algebra/calculus to solve for unknown values.

- ESP32 NodeMCU Development Board 38-pin Type-C:
 - the "brain" of the device. will process firmware, run equations, store database arrays and coordinate I/O.
- 2.8" TFT LCD ILI9341 SPI:
 - visual output screen. renders ui, math formulas, menus, and calculation results.
- 6x6x5 tactile push button switch: x 20-x40
 - manual input controls, used for: navigating menus, scrolling through formula lists, and typing in values.
- TP4056 1A LiPo Battery Charger Module:
 - allows for recharging of battery without risk of overcharging
- 3.7V 1200 mAh LiPo Battery:
 - portable power source, provides clean power
- 1N4148 Switching Diodes: x 50
 - prevents key ghosting



Menu Screens

- ▷ 1. DC & AC Circuits 2. Semiconductors & Diodes
- ▷ 3. Filters, Op-Amps and Transients 4. Mechanics & Physics
- ▷ 5. Calculus & Math tools 6. Utilities & Constants
- ▷ 1. Ohm's Law, $V=IR$ | Power, $P=VI=I^2R=\frac{V^2}{R}$
- ▷ Resistor Series, $R_{total}=R_1+R_2...$ | Resistor Parallel, $\frac{1}{R_{total}}=\frac{1}{R_1}+\frac{1}{R_2}...$
- ▷ Voltage divider, $V_{out}=V_{in}\left(\frac{R_2}{R_1+R_2}\right)$ | AC AM & Peak, $V_{rms}=\frac{V_{peak}}{\sqrt{2}} \approx 0.707 \cdot V_{peak}$
- ▷ Reactance, $X_c=\frac{1}{2\pi fC}$ and $X_L=2\pi fL$

- ▷ 2. Shockley Diode, $I_D=I_s\left(e^{\frac{V_D}{nV_T}}-1\right)$ | Zener Reg. Ser. Res., $R_s=\frac{V_{in}-V_z}{I_z+I_{load}}$
- ▷ Full-Wave Rectifier DC Output: $V_{dc}=\frac{2V_{peak}}{\pi}$

- ▷ 3. Filters, Op-Amps, and Transients
- ▷ RC Cutoff Freq. (-3 dB point): $f_c=\frac{1}{2\pi RC}$ | Active Op-Amp Voltage Gain
- ▷ Non inverting: $A_v=1+\frac{R_f}{R_{in}}$ | Inverting: $A_v=-\frac{R_f}{R_{in}}$ | LC Tank
- ▷ resonant freq: $f_0=\frac{1}{2\pi\sqrt{LC}}$

- ▷ 4. Calc based mech & Math
- ▷ Trapezoidal integration loop: $\int_a^b f(x)dx \approx \frac{\Delta x}{2} \sum_{i=1}^n (f(x_{i-1})+f(x_i))$
- ▷ Work by variable force $W=\int_{x_1}^{x_2} F(x)dx$ | Kinematics via
- ▷ integration: $v(t)=v_0+\int_{t_0}^t a(t)dt$ and $x(t)=x_0+\int_{t_0}^t v(t)dt$

- ▷ 5. Definite integral: $\int_a^b f(x)dx \approx \frac{\Delta x}{2} [f(x_0)+2\sum_{i=1}^{n-1} f(x_i)+f(x_n)]$
- ▷ Average Val of Fxn: $f_{avg}=\frac{1}{b-a} \int_a^b f(x)dx$
- ▷ Right triangle solver $c=\sqrt{a^2+b^2}$; SOH CAH TOA

- ▷ 6. Units & Prefixes
- ▷ Tera: 10^{12} Giga: 10^9 Mega: 10^6 Kilo: 10^3 Milli: 10^{-3} Micro: 10^{-6}
- ▷ Nano: 10^{-9} Pico: 10^{-12} SOL: $3 \times 10^8 \frac{m}{s}$ Grav. Accel.: $9.81 \frac{m}{s^2}$
- ▷ PoFS: $8.854 \times 10^{-12} \frac{F}{m}$ Elem. Charge: $1.602 \times 10^{-19} C$
- ▷ Planck's Constant: $6.626 \times 10^{-34} J \cdot s$